

Chemical Kinetics

NCERT Class XII Chemistry · Chapter 3 · Chemical Kinetics

Chemical Kinetics — Complete Notes

13 marks. Almost pure numerical territory — order from data, integrated rate equations, half-life, activation energy. Four formulas do nearly all the work; the marks go to knowing which one the question is pointing at.

What kinetics answers — *and what it deliberately doesn't*

Three different questions about a reaction, three different branches:

Question	Answered by
Is it feasible? ($\Delta G < 0$)	Thermodynamics
How far does it go?	Chemical equilibrium (K)
How fast, and by what mechanism?	Chemical kinetics

The standard illustration: thermodynamics says diamond $\rightarrow$ graphite is feasible. Kinetics explains why your ring is safe — the conversion is immeasurably slow.

Rate of reaction — *average, instantaneous, and the coefficient division*

Rate = change in concentration per unit time. Reactant concentration falls, so its rate carries a minus sign to keep the rate positive:

$$\text{rate} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$

Average rate uses Δ over a finite interval. **Instantaneous rate** is the slope of the tangent at one moment: $-\frac{d[R]}{dt}$.

When coefficients differ — divide by them

For $aA + bB \rightarrow cC + dD$:

$$\text{rate} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

Trap: "rate of reaction" and "rate of disappearance of B" are different numbers whenever B's coefficient isn't 1. Read which one is asked — this is the entire content of a 2-marker that runs most years.

Worked · 2022 — rate of reaction vs rate of disappearance

$\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$, rate of formation of $\text{NH}_3 = 3.6 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. Find the rate of reaction and the rate of disappearance of H_2 .

$$\text{rate} = \frac{1}{2} \frac{\Delta[\text{NH}_3]}{\Delta t} = \frac{1}{2}(3.6 \times 10^{-4}) = 1.8 \times 10^{-4}$$

$$-\frac{\Delta[\text{H}_2]}{\Delta t} = 3 \times \text{rate} = 5.4 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

Examiner asks: this exact shape, with $\text{N}_2/\text{H}_2/\text{NH}_3$ or $A + 3B \rightarrow 2C$. Near-guaranteed 2-3 marker.

Rate law and finding the order from data — *the standard 3-marker, step by step*

The rate law is **experimental** — powers are not read off the balanced equation:

$$\text{rate} = k[A]^x[B]^y$$

x and y may or may not equal the stoichiometric coefficients. **Order** = $x + y$. Order can be zero, fractional, or negative; molecularity cannot.

Getting x and y from a data table — the method

1. Pick two experiments where **one** concentration is held constant and the other changes.
2. Write the rate law for each; divide one by the other. k and the constant term cancel.
3. Read off the power that makes the ratio work.
4. Repeat with a different pair for the other exponent.
5. Substitute any single experiment back to find k .

Trap: choosing two experiments where *both* concentrations change. Nothing cancels and the algebra becomes unsolvable. Scan the table for the constant column first.

Order vs molecularity — the comparison they ask for

Order	Molecularity
Sum of powers in the experimental rate law	Number of species colliding in one elementary step
Experimental	Theoretical
Can be 0, fractional, negative	Whole number only (1, 2, 3)
Applies to overall reaction	Only defined for an elementary step

Examiner asks: "can this be an elementary reaction?" (2026) with a fractional order like $3/2$ — **no**, because an elementary reaction's order equals its molecularity, which must be a whole number.

Units of the rate constant — *one formula, every order*

From $\text{rate} = k[\]^n$, so $k = \text{rate}/[\]^n$:

$$\text{unit of } k = \text{mol}^{1-n} \text{ L}^{n-1} \text{ s}^{-1}$$

Order	Unit of k
Zero	$\text{mol L}^{-1} \text{ s}^{-1}$
First	s^{-1}
Second	$\text{L mol}^{-1} \text{ s}^{-1}$

Examiner asks: given a rate law, state the units — or, run backwards, given the units, state the order. Both directions appear.

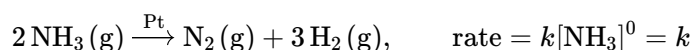
Integrated rate equations — *zero and first order, plus their graphs*

Zero order

$$[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$$

Plot $[R]$ against t : straight line, **slope** = $-k$, intercept $[R]_0$.

Example: decomposition of gaseous ammonia on a hot platinum surface at 1130 K. At high pressure the metal surface saturates with ammonia, so adding more changes nothing — rate becomes independent of concentration:



First order

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

Plot $\log \frac{[R]_0}{[R]}$ against t : straight line through the origin, **slope** = $k/2.303$. Plot $\ln[R]$ against t instead and the slope is $-k$, intercept $\ln[R]_0$.

Examples: hydrogenation of ethene; all natural and artificial radioactive decay; decomposition of N_2O_5 and N_2O .

First order in the gas phase — the closed form worth memorising

Concentration is proportional to partial pressure, so the same equation runs on pressures. For $A(\text{g}) \rightarrow B(\text{g}) + C(\text{g})$ with all coefficients 1, let x be the pressure decomposed:

$$p_{\text{total}} = (p_i - x) + x + x = p_i + x \Rightarrow x = p_T - p_i$$

So the reactant's partial pressure is $p_A = p_i - x = 2p_i - p_T$, and:

$$k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_T}$$

Use this directly whenever the question gives you **initial and total** pressure. If it gives the reactant's partial pressure outright, the plain $\log(p_i/p_A)$ form is enough.

Worked · 2016 — rate constant from total pressure

$\text{C}_2\text{H}_5\text{Cl}(\text{g}) \longrightarrow \text{C}_2\text{H}_4(\text{g}) + \text{HCl}(\text{g})$. At $t = 0$, $p_i = 0.30$ atm; at $t = 300$ s, $p_T = 0.50$ atm. ($\log 3 = 0.4771$)

$$k = \frac{2.303}{300} \log \frac{0.30}{2(0.30) - 0.50} = \frac{2.303}{300} \log \frac{0.30}{0.10}$$

$$= \frac{2.303 \times 0.4771}{300} = 3.66 \times 10^{-3} \text{ s}^{-1}$$

The unit is s^{-1} , which itself confirms first order.

Trap: feeding the *total* pressure straight into the log. It must be $2p_i - p_T$, not p_T .

Half-life — and the graph question that turns on it

Time for half the reactant to be consumed.

$$\text{First order: } t_{1/2} = \frac{0.693}{k} \quad \text{Zero order: } t_{1/2} = \frac{[R]_0}{2k}$$

The distinction that gets examined: first-order half-life is **independent of initial concentration**; zero-order half-life is **directly proportional** to it.

Examiner asks: "predict the order from the graph" (2019) — a flat $t_{1/2}$ vs $[R]_0$ line means first order; a straight rising line means zero order.

Percentage-completion problems

Set $[R]_0 = 100$ and subtract the percentage completed. 75% done $\rightarrow [R] = 25$; 99% done $\rightarrow [R] = 1$.

Worked · 2026 — time for 3/4 decomposition

First order, $k = 2.54 \times 10^{-3} \text{ s}^{-1}$. Time for 3/4 of the reactant to decompose? ($\log 4 = 0.6$)

$[R]_0 = a$, so $[R] = a - \frac{3}{4}a = \frac{a}{4}$.

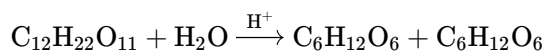
$$t = \frac{2.303}{k} \log \frac{a}{a/4} = \frac{2.303 \times 0.6}{2.54 \times 10^{-3}} = 544 \text{ s}$$

A neat consequence worth remembering: for first order, $t_{99\%} = 2 \times t_{90\%}$ — because $\log 100 = 2$ and $\log 10 = 1$.

Pseudo first order — *two examples, one idea*

A reaction that is **bimolecular but follows first-order kinetics**, because one reactant is present in such large excess that its concentration doesn't measurably change.

Hydrolysis of sucrose:



Water is the solvent, so $[\text{H}_2\text{O}]$ is effectively constant. Rate = $k[\text{C}_{12}\text{H}_{22}\text{O}_{11}]$: order 1, molecularity 2.

Hydrolysis of an ester in dilute acid behaves the same way.

Examiner asks: (2024) write the rate law, then state order *and* molecularity separately, then name the reaction type. The gap between order 1 and molecularity 2 is the whole point.

Temperature dependence and Arrhenius — *the 5-marker's favourite*

Rate roughly **doubles for every 10 K rise** near room temperature. Quantitatively:

$$k = A e^{-E_a/RT}$$

A = frequency factor, E_a = activation energy, the minimum energy colliding molecules need.

The logarithmic form and its graph

$$\ln k = -\frac{E_a}{RT} + \ln A$$

Against $y = mx + c$ with $y = \ln k$ and $x = 1/T$:

- **Slope** = $-E_a/R$ (negative, so the line falls)
- **Intercept** = $\ln A$

Two-temperature form — the one used in numericals

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Worked · 2025 — E_a when the rate doubles over 10 K

Rate doubles from 298 K to 308 K. $2.303R = 19.15$, $\log 2 = 0.3$.

Concentrations are unchanged, so a doubled rate means a doubled k : $k_2/k_1 = 2$.

$$0.3 = \frac{E_a}{19.15} \left[\frac{10}{308 \times 298} \right]$$

$$E_a = \frac{0.3 \times 19.15 \times 308 \times 298}{10} = 52729.9 \text{ J mol}^{-1} = 52.7 \text{ kJ mol}^{-1}$$

Worked · 2025 — E_a from two half-lives

First order, 50% complete in 20 min at 300 K and in 5 min at 350 K. ($\log 4 = 0.602$)

50% completion is the half-life, so $k = 0.693/t_{1/2}$:

$$\frac{k_2}{k_1} = \frac{0.693/5}{0.693/20} = 4$$

$$0.602 = \frac{E_a}{2.303 \times 8.314} \left[\frac{50}{300 \times 350} \right] \Rightarrow E_a = 24205 \text{ J mol}^{-1} = 24.2 \text{ kJ mol}^{-1}$$

Trap: "50% completed in 20 minutes" is a half-life in disguise. Spot that and the question collapses into two lines.

Collision theory and catalysis — *the definitions that carry marks*

Activated complex and the energy profile

Colliding reactant molecules first form a short-lived **activated complex**. The energy needed to reach it is the **activation energy**. The reaction's own energy change is separate:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

On a potential-energy vs reaction-coordinate plot, the barrier height is E_a either way; what changes is where the products land. Products lower than reactants $\rightarrow \Delta H$ negative $\rightarrow$ **exothermic**. Products higher $\rightarrow \Delta H$ positive $\rightarrow$ **endothermic**.

Maxwell-Boltzmann distribution

Plot the fraction of molecules (n_E/n_T) against kinetic energy. Most molecules sit near the **most probable kinetic energy**; only a small tail — roughly 10-20% — carries energy above E_a , and only those can react.

Raise the temperature by 10 K and the curve flattens and shifts right: the fraction with energy at or above E_a roughly **doubles**. That is precisely why the rate doubles for a 10 K rise.

Collision theory

Treats molecules as hard spheres that must collide to react.

Collision frequency (Z): number of collisions per second per unit volume of reaction mixture.

$$\text{rate} = Z_{AB} e^{-E_a/RT}$$

Comparing this with Arrhenius shows the frequency factor A is essentially the **collision frequency**. The theory is accurate for atoms and simple molecules but deviates for complex ones — because not every collision works. Two conditions must both hold:

1. Colliding molecules have energy at least equal to the **threshold energy**.
2. They collide in the **proper orientation**, so old bonds can break and new ones form.

Catalysis

A catalyst increases the rate **without itself undergoing any permanent chemical change**. By **intermediate complex theory**, it forms a temporary bond with the reactant, making a transitory intermediate that decomposes into product and releases the catalyst again — which is why a small amount catalyses a large amount of reactant.

The effect is an **alternative pathway of lower activation energy**, so more molecules clear the barrier and the rate rises.

What a catalyst does not change: the Gibbs energy of the reaction, the equilibrium constant, or the position of equilibrium. It catalyses forward and backward reactions *to the same extent*, so equilibrium is reached **earlier** but at the same place. And since it can't change ΔG , it cannot make a non-spontaneous reaction happen — only speed up one that already is.

Examiner asks: (2017) "effect of a catalyst on activation energy and on ΔG " — lowers E_a , leaves ΔG untouched. Answering only the first half loses the mark.

Reaction mechanism and the rate-determining step

For a multi-step reaction, the **slowest step determines the rate** — write the rate law directly from that step's reactants.

Example (2025): $2\text{H}_2\text{O}_2 \xrightarrow{\text{I}^-} 2\text{H}_2\text{O} + \text{O}_2$ via a slow step $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$ and a fast step. So rate = $k[\text{H}_2\text{O}_2][\text{I}^-]$, overall order 2 — even though the balanced equation shows $2\text{H}_2\text{O}_2$.

Numerical patterns, collected — *five patterns, one model each*

A · Rate from stoichiometry — 2 marks

Recognise it: one species' rate is given, another is asked for.

1. Write the full rate expression with 1/coefficient on every term.
2. Equate the known term to the wanted one.
3. Check whether "rate of reaction" or "rate of disappearance" is asked.

B · Order from a data table — 3 marks

Recognise it: a table of concentrations and initial rates.

1. Find two rows with one concentration constant.
2. Divide the two rate laws; solve for the exponent.
3. Repeat for the other exponent, then back-substitute for k .

C · Integrated first-order equation — 3 marks

Recognise it: a percentage decomposed, or concentrations at two times.

1. Set $[R]_0 = 100$ and subtract the percentage.
2. $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$, rearranged for the unknown.
3. Gas phase $\rightarrow$ convert total pressure to reactant partial pressure first.

D · Half-life $\leftrightarrow$ rate constant — 2–3 marks

Recognise it: "half-life", or "50% completed".

1. Identify the order from the rate law or the units of k .
2. First order: $t_{1/2} = 0.693/k$. Zero order: $t_{1/2} = [R]_0/2k$.

Worked · 2026 — half-life from percentage decomposition

First order, 25% decomposed in 25 min. Find $t_{1/2}$. ($\log 4 = 0.6021$, $\log 3 = 0.4771$)

$$k = \frac{2.303}{25} \log \frac{100}{75} = \frac{2.303}{25} (0.6021 - 0.4771) = 0.0115 \text{ min}^{-1}$$

$$t_{1/2} = \frac{0.693}{0.0115} = 60.26 \text{ min}$$

E · Activation energy from two temperatures — 5 marks

Recognise it: two temperatures, or "rate doubles when temperature rises by..."

1. Get k_2/k_1 — from rates directly, or from $0.693/t_{1/2}$ at each temperature.
2. $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$.
3. Answer in J mol⁻¹, then divide by 1000 for kJ.

Past year questions · question types — *ranked by how often they turn up*

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · First-order integrated rate equation — 3 marks

Recognise it: a percentage decomposed, or a concentration/pressure at two times.

1. $[R]_0 = 100$, subtract the percentage.
2. Apply the 2.303 formula.
3. Convert total pressure to partial pressure if gaseous.

Trap: using the percentage *decomposed* as $[R]$ instead of what remains.

2 · Activation energy (Arrhenius) — 5 marks

Recognise it: two temperatures, or $2.303R = 19.15$ handed to you.

1. Get k_2/k_1 .
2. Two-temperature Arrhenius form.
3. Convert to kJ if the answer is large.

Trap: inverting $T_2 - T_1$ over $T_1 T_2$, or forgetting that the half-life ratio inverts the k ratio.

3 · Order from a data table — 3-5 marks

Recognise it: a table of initial concentrations and rates.

1. Pair rows sharing a constant concentration.
2. Divide; solve the exponent.
3. Back-substitute for k and state its units.

4 · Rate expression from stoichiometry — 2-3 marks

Recognise it: one species' rate given, another asked.

1. Full expression with 1/coefficient terms.
2. Equate and solve.

5 · Definitions and comparisons — 2 marks

Recognise it: "define", "distinguish between", "write the unit of".

1. Order vs molecularity: answer in matched pairs.
2. Half-life, collision frequency, effective collision, activation energy — book definitions.

6 · Graph interpretation — 3 marks

Recognise it: a plot of $\ln k$ vs $1/T$, or $t_{1/2}$ vs $[R]_0$, or $\log \frac{[R]_0}{[R]}$ vs t .

1. Match the plot to its equation.
2. Read slope and intercept off $y = mx + c$.
3. State what each represents in symbols, not words alone.

Past year questions · mark slots — *what each type is worth*

Question type	Slot	Time
First-order integrated equation	3-marker	4 min
Activation energy	5-marker (or part of one)	6-7 min
Order from data table	3-marker, sometimes a 5	5-6 min
Rate from stoichiometry	2-3 marker	3 min
Definitions / comparisons	2-marker	2 min
Graph interpretation	3-marker	4 min
Pseudo first order	3-marker	3 min

The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — activation energy and the multi-part data questions are where the 5-markers sit.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2018 Q15 · lecture PYQ

"Second order in A, first order in B" — write the differential rate equation, then how the rate changes when concentrations are tripled or doubled. Identical wording both times.

2025 Q16 · 2025 Q23(a)

Activation energy from two temperatures — once from two half-lives, once from a rate that doubles over 10 K. Same formula, two disguises.

2019 Q5 · 2022 Q17

Graph questions: $\ln k$ vs $1/T$ (intercept and slope), and $\log \frac{[R]_0}{[R]}$ vs t (order and slope). Both reduce to $y = mx + c$.

2022 Q8 · 2026 Q6

Order vs molecularity — as a straight comparison, and as "can this be an elementary reaction" with a fractional order.

2024 Q21 · lecture PYQ

Sucrose hydrolysis: rate law, order 1 but molecularity 2, and the name — pseudo first order.

2019 Q9 · 2019 Q10

Half-life: predicting order from a $t_{1/2}$ vs $[R]_0$ graph, and computing $t_{1/2}$ from a first-order rate law.

2026 Q1 · 2024 Q3

Definitions: collision frequency, effective collisions, half-life period, and the units of k for zero and first order.

Past year questions · cold practice — *answers only* — *work them before you look*

Integrated rate equation

- 2017 Q11 — N_2O_5 decomposition: 1.6×10^{-2} M at $t = 0$, 0.8×10^{-2} M at 300 s. Show first order and find $t_{1/2}$. $k = 2.3 \times 10^{-3} \text{ s}^{-1}$ at both times; $t_{1/2} = 301 \text{ s}$
- 2026 Q22 — $\text{C}_2\text{H}_5\text{Cl}$: $p_i = 0.30 \text{ atm}$, total $p = 0.50 \text{ atm}$ at 30 s. Find k . 0.0368 s^{-1}
- 2026 Q4 — $k = 2.54 \times 10^{-3} \text{ s}^{-1}$, time for 3/4 decomposition. 544 s
- 2024 — first order, $k = 1.25 \times 10^{-3} \text{ s}^{-1}$. Time for 5 g to fall to 2.5 g. 554.6 s — note this is just $t_{1/2}$ in disguise
- 2024 — show $t_{99\%} = 2 t_{90\%}$ for first order. $\log 100 = 2$ vs $\log 10 = 1$
- 2021-22 — first order, 75% decomposed in 30 min. Find $t_{1/2}$. $k = 0.046 \text{ min}^{-1}$; $t_{1/2} = 15 \text{ min}$
- Lecture PYQ — zero order, $[R]_0 = 0.1 \text{ M}$ falls to 0.064 M, $k = 4 \times 10^{-3}$. Find t . 9 s

Half-life

- 2019 Q10 — rate $= 5.5 \times 10^{-14} [\text{C}_2\text{H}_4]$. Units of k and $t_{1/2}$. s^{-1} ; $1.26 \times 10^{13} \text{ s}$
- 2026 Q24(b) — first order, 25% decomposed in 25 min. Find $t_{1/2}$. $k = 0.0115 \text{ min}^{-1}$; $t_{1/2} = 60.26 \text{ min}$

Activation energy

- 2025 Q16 — 50% complete in 20 min at 300 K, 5 min at 350 K. Find E_a . 24.2 kJ mol^{-1}

- 2025 Q23(a) — rate doubles from 298 K to 308 K. Find E_a . 52.7 kJ mol^{-1}

Rate and order

- 2020 — $A + 3B \rightarrow 2C$, rate of formation of C = 2.5×10^{-4} . Find rate of reaction and rate of disappearance of B. 1.25×10^{-4} ; $3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- 2026 Q6 — rate = $k[A][B]^{3/2}$. Overall order, and can it be elementary? 2.5; no — elementary reactions have whole-number order
- 2025 Q18 — order with respect to A and B from a rate table where doubling B leaves the rate unchanged. Order in B is zero

Mechanism

- 2025 Q23(b) — $2\text{H}_2\text{O}_2 \xrightarrow{\text{I}^-} 2\text{H}_2\text{O} + \text{O}_2$, slow step $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$. Rate law and overall order. rate = $k[\text{H}_2\text{O}_2][\text{I}^-]$; order 2

Built from Sourabh Raina's Chemical Kinetics one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 3 (Rationalised 2022-23). NCERT confirms the Arrhenius form $k = Ae^{-E_a/RT}$ and the term "pseudo first order reaction".